



Graphs in Machine Learning

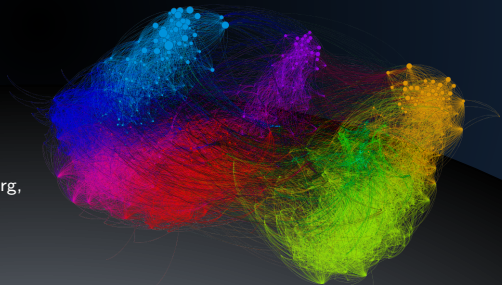
Spectral Clustering: Relaxation

From Discrete to Continuous

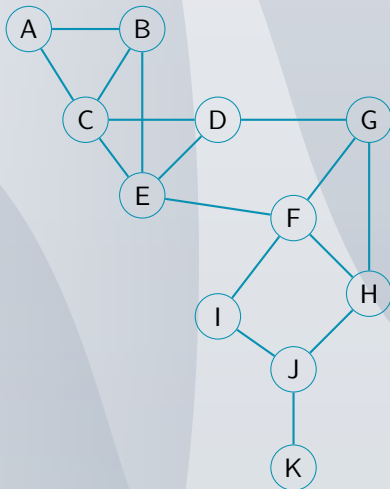
Michal Valko

Isara Labs

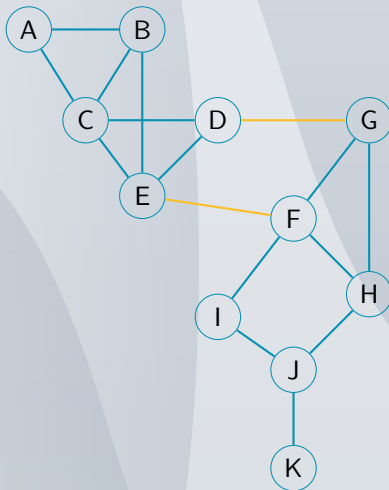
Partially based on material by: Ulrike von Luxburg,
Gary Miller, Doyle & Schnell, Daniel Spielman



Spectral Clustering: Cuts on graphs

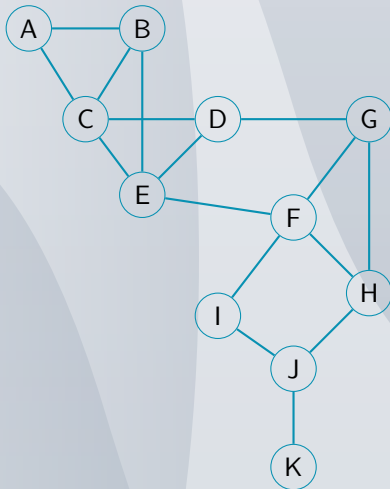


Spectral Clustering: Cuts on graphs



Defining the cut objective we get the clustering!

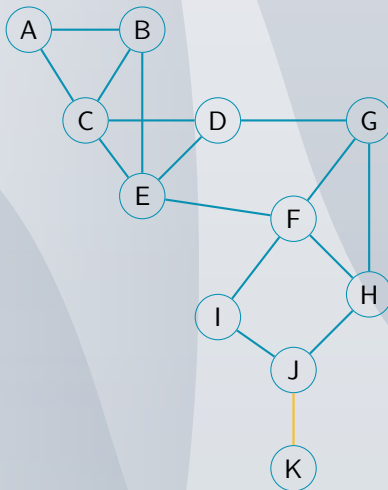
Spectral Clustering: Cuts on graphs



MinCut: $\text{cut}(A, B) = \sum_{i \in A, j \in B} w_{ij}$

Are we done?

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Are we done?

Can be solved efficiently, but maybe not what we want

Spectral Clustering: Balanced Cuts

Let's balance the cuts!

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RatioCut

$$\text{RatioCut}(A, B) = \sum_{i \in A, j \in B} w_{ij} \left(\frac{1}{|A|} + \frac{1}{|B|} \right)$$

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Normalized Cut

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Easily generalizable to $k \geq 2$

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Approximate!

Spectral Clustering: Relaxing Balanced Cuts

Relaxation for (simple) balanced cuts for 2 sets

$$\min_{A,B} \text{cut}(A, B) \quad \text{s.t.} \quad |A| = |B|$$

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Graph function $\mathbf{f}$ for cluster membership

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$$\text{cut}(A, B) = \sum_{i \in A, j \in B} w_{i,j} = \frac{1}{4} \sum_{i,j} w_{i,j} (f_i - f_j)^2 = \frac{1}{2} \mathbf{f}^T \mathbf{L} \mathbf{f}$$

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What is the relationship with the **smoothness** of a graph function?

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$$\|\mathbf{f}\| =$$

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objective function of spectral clustering

$$\min_{\mathbf{f}} \mathbf{f}^T \mathbf{L} \mathbf{f} \quad \text{s.t.} \quad f_i = \pm 1, \quad \mathbf{f} \perp \mathbf{1}_N, \quad \|\mathbf{f}\| = \sqrt{N}$$

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$$\cancel{f_i = \pm 1} \rightarrow f_i \in \mathbb{R}$$

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Rayleigh-Ritz theorem

If $\lambda_1 \leq \dots \leq \lambda_N$ are the eigenvalues of real symmetric $\mathbf{L}$ then

$$\lambda_1 = \min_{\mathbf{x} \neq 0} \frac{\mathbf{x}^T \mathbf{L} \mathbf{x}}{\mathbf{x}^T \mathbf{x}} = \min_{\mathbf{x}^T \mathbf{x} = 1} \mathbf{x}^T \mathbf{L} \mathbf{x}$$
$$\lambda_N = \max_{\mathbf{x} \neq 0} \frac{\mathbf{x}^T \mathbf{L} \mathbf{x}}{\mathbf{x}^T \mathbf{x}} = \max_{\mathbf{x}^T \mathbf{x} = 1} \mathbf{x}^T \mathbf{L} \mathbf{x}$$

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How can we use it?

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Generalized Rayleigh-Ritz theorem (Courant-Fischer-Weyl)

If $\lambda_1 \leq \dots \leq \lambda_N$ are the eigenvalues of real symmetric $\mathbf{L}$ and $\mathbf{v}_1, \dots, \mathbf{v}_N$ the corresponding orthogonal eigenvectors, then for $k = 1 : N - 1$

$$\lambda_{k+1} = \min_{\mathbf{x} \neq 0, \mathbf{x} \perp \mathbf{v}_1, \dots, \mathbf{v}_k} \frac{\mathbf{x}^T \mathbf{L} \mathbf{x}}{\mathbf{x}^T \mathbf{x}} = \min_{\mathbf{x}^T \mathbf{x} = 1, \mathbf{x} \perp \mathbf{v}_1, \dots, \mathbf{v}_k} \mathbf{x}^T \mathbf{L} \mathbf{x}$$

$$\lambda_{N-k} = \max_{\mathbf{x} \neq 0, \mathbf{x} \perp \mathbf{v}_N, \dots, \mathbf{v}_{N-k+1}} \frac{\mathbf{x}^T \mathbf{L} \mathbf{x}}{\mathbf{x}^T \mathbf{x}} = \max_{\mathbf{x}^T \mathbf{x} = 1, \mathbf{x} \perp \mathbf{v}_N, \dots, \mathbf{v}_{N-k+1}} \mathbf{x}^T \mathbf{L} \mathbf{x}$$

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<https://misovalko.github.io/mva-ml-graphs.html>